

**Jeffrey D. Walker**  
Yale University

100 College St. Rm 1126  
New Haven, CT 06510

917-670-8073  
jeffrey.walker@yale.edu

**CURRENT POSITION**

Associate Research Scientist, Saxena Lab  
Yale University, Department of Biomedical Engineering  
Wu Tsai Institute, Center for Neurocomputation and Machine Intelligence

October 2024 – present

**EDUCATION**

**University of Chicago**

PhD, Computational Neuroscience

Chicago, IL  
October 2012 - July 2018

Dissertation: Methods for studying the neural control of natural behaviors using unconstrained marmosets

Advisor: Nicholas Hatsopoulos, Professor Organismal Biology & Anatomy

**Wesleyan University**

MA, Neuroscience and Behavior

Middletown, CT  
2008 – 2009

Thesis: The Propagation of epileptiform events across the corpus callosum

Advisor: Gloster Aaron, Associate Professor Neuroscience and Behavior

BA, Honors, Neuroscience and Behavior

2004 – 2008

Thesis: The Interface of phenomenology and the cognitive sciences

Advisor: Steve Horst, Professor of Philosophy

**RESEARCH EXPERIENCE**

**University of Chicago, Organismal Biology & Anatomy**

Postdoctoral Scholar; Advisors: Nicholas Hatsopoulos and Jason MacLean

2018 – 2024

Sensorimotor cortical population dynamics across marmoset natural behaviors and sleep

**University of Chicago, Committee on Computational Neuroscience**

Graduate Research Assistant; Advisor: Nicholas Hatsopoulos

2012 – 2018

Methods to study the neural control of movement with common marmosets

**Wesleyan University, Biology Department**

Student then lab technician; Advisor: Gloster Aaron

2007 – 2010

Propagation of epileptiform events across the corpus callosum

**UC Davis Center for Mind and Brain**

Intern; Supervisor: Cliff Saron

Summer 2006

**PUBLICATIONS**

†:co-senior author, \*:co-first author

Muhammad Noman Almani\*, John Lazzari\*, **J Walker\***, Shreya Saxena (2026) Embodied sensorimotor control: computational modeling of the neural control of movement. *Annual Reviews in Biomedical Engineering*. 10.1146/annurev-bioeng-102723-020454.

Paul Aparicio\*, **Jeffrey D. Walker\***, Jason N. MacLean, Nicholas G. Hatsopoulos (2025) Spatially extensive LFP correlations identify slow-wave sleep in marmoset sensorimotor cortex. *eNeuro*. 10.1523/ENEURO.0139-25.2025

- Moore D, MacLean J, **Walker J**<sup>†</sup>, Hatsopoulos N<sup>†</sup> (2024) A dynamic subset of network interactions underlies tuning to natural movements in marmoset sensorimotor cortex. *Cell Reports*. 15:10517
- Moore D, **Walker J**, MacLean J, Hatsopoulos N (2022) Validating markerless pose estimation with 3D X-ray radiography. *Journal of Experimental Biology*. 225, jeb243998. 10.1242/jeb.243998.
- Walker J**, Pirschel F, Sundiang M, Niekrasz M, MacLean J, Hatsopoulos N (2021) Chronic wireless neural population recordings with common marmosets. *Cell Reports* 36(2):109379. 10.1016/j.celrep.2021.109379.
- Walker J**, Pirschel F, Gidmark N, MacLean J, Hatsopoulos N (2020) A platform for semi-automated voluntary training of common marmosets for behavioral neuroscience. *Journal of Neurophysiology* 123(4):1420-1426. 10.1152/jn.00300.2019.
- Walker J**, MacLean J, Hatsopoulos N (2017) The marmoset as a model system for studying voluntary motor control. *Developmental Neurobiology*. 77(3):273-285. 10.1002/dneu.22461
- Walker J**, Storch G, Quach-Wong B, Sonnenfeld J, Aaron G (2012) Propagation of epileptiform events across the corpus callosum in a cortical slice preparation. *PLOS ONE*. 7(2):e31415. 10.1371/journal.pone.0031415
- Aaron G, Briggs S, **Walker J**, Asik K, Lombroso P, Naegel J (2011) STEP Regulation of seizure thresholds in the hippocampus. *Epilepsia* 52(3):497-506.

## PUBLICATIONS IN PREPARATION

- Walker J**, Aparicio P, Saxena S, Hatsopoulos N, MacLean J (2026) Feedback-driven sensorimotor cortical dynamics during natural prey capture.
- Walker J**, MacLean J, Hatsopoulos N, Saxena S (2026) Dynamics of Sensorimotor Control in a Virtual Marmoset.

## CONFERENCE PRESENTATIONS

- Walker J**, MacLean J, Hatsopoulos N, Saxena S (2026) Embodied simulation of natural behavior. *Simulated Bodies: Whole-body biomechanical models*. Janelia Research Campus. Ashburn, Virginia.
- Yongxu Z, **Walker J**, Hatsopoulos N, MacLean J, Saxena S (2026) Uncovering Multi-Regional Inputs Using Input-Driven Switching Recurrent Neural Networks. CoSyNe Lisbon, Portugal.
- Samantha Johnson, **Walker J**, Hatsopoulos N (2026) Network features underlying motor learning and skill generalization in marmosets. Neural Control of Movement. Kobe, Japan
- Walker J**, MacLean J, Hatsopoulos N, Saxena S (2025) Embodied simulation of natural behavior. *Neurocybernetics at Scale*. Champalimaud Research Symposium. Champalimaud Center for the Unknown, Lisbon.
- Walker J**, MacLean J, Hatsopoulos N (2024) Building a whole -body marmoset model for deep reinforcement learning driven musculoskeletal simulation to understand sensorimotor control. *Simian Collective*. Pittsburgh.
- Walker J**, Hatsopoulos N (2023) Building a whole-body marmoset model for deep reinforcement learning driven musculoskeletal simulation to understand sensorimotor control. *Simulated Bodies: Whole body biomechanical models*. Janelia Research Campus, Ashburn, Virginia.
- Walker J**, Paul Aparicio, Jason N. Maclean, Nicholas G. Hatsopoulos. (2023) Toward understanding the dynamics of feedback control during prey capture. *Marmoset Biosciences Symposium*. Washington D.C.

- Moore D\*, **Walker J\***, Aparicio P, Tortolani A, MacLean J, Hatsopoulos N (2023) A sensorimotor cortical encoding model incorporating kinematic trajectories and network effects during natural forelimb movements of the marmoset. *Simian Collective*. Chicago.
- Walker J**, Moore D, MacLean J, Hatsopoulos N (2022) Understanding sensorimotor control through marmoset prey capture. *Society for Neuroscience Annual Meeting*. San Diego.
- Aparicio P, **Walker J**, Tortolani A, Hatsopoulos N (2022) 2D Spatial structure in correlations across sensorimotor cortex of freely behaving marmosets. *Society for Neuroscience Annual Meeting*. San Diego.
- Moore D\*, **Walker J\***, Hatsopoulos N, MacLean J (2022) Developing a network encoding model for natural behavior in the motor cortex of the unconstrained marmoset. *ARIADNE*, Santorini, Greece.
- Walker J**, MacLean J, Hatsopoulos N (2021) Characterizing neuromechanical strategies used by marmosets during foraging and prey capture. *Society for Neuroscience*. Online.
- Walker J**, Bachschmid Romano L, Moore D, Aparicio P, Sundiang M, Brunel N, Hatsopoulos N, MacLean J (2021) The importance of spatially dependent neuronal interactions in modeling cortical activity during unconstrained ethological motor behaviors in the marmoset. NIH Brain Initiative Investigators meeting. Online
- Walker J**, Moore D, Sundiang M, MacLean J, Hatsopoulos N (2019) Chronic wireless neural population recordings in freely behaving marmosets. *Society for Neuroscience Annual Meeting*. Chicago.
- Sundiang M, **Walker J**, Moore D, MacLean J, Hatsopoulos N (2019) Developing a motor task platform for freely-behaving common marmosets. *Society for Neuroscience Annual Meeting*. Chicago.
- Moore D, **Walker J**, MacLean M, Hatsopoulos N (2019) Comparing pose estimation performance using DeepLabCut and marker-based tracking. *Society for Neuroscience Annual Meeting*. Chicago.
- Walker J**, Pirschel F, MacLean J, Hatsopoulos N (2018) Encoding spaces across the behavioral repertoire of the common marmoset. *ARIADNE*, Santorini, Greece.
- Walker J**, Pirschel F, MacLean J, Hatsopoulos N (2018) Sensorimotor cortical population responses across natural behaviors. *Society for Neuroscience Annual Meeting*. San Diego.
- Walker J**, Hatsopoulos N (2015) Design of an experimental platform for voluntary behavioral training and neurophysiological study of the common marmoset. *Society for Neuroscience Annual Meeting*. Chicago.
- Walker J**, Leung S, Aaron G (2009) The Propagation of epileptiform events across the corpus callosum: results from an in vitro preparation. *American Epilepsy Society*.
- Walker J** (2007) Does inhibition increase spike time precision? *Howard Hughes Summer Research Program*.
- Walker J** (2006) The Interface of science and Buddhism: will it alter a tradition? *Wesleyan Center for East Asian Studies*.

## INVITED TALKS

Inaugural Chicago Symposium on Computational Neuroscience  
IBM, Health Care Life Science Research

June 2022  
October 2021

## SERVICE

Pathways to Science (YPRI), mentor, Yale University

Summer 2026  
Walker 3

Served as mentor to New Haven high school student doing summer internship in science lab at Yale.

**Initiative for Maximizing Student Development (IMSD)**, peer mentor, University of Chicago 2015 – 2016  
Participated in monthly discussion group around maximizing career success among graduate students from underrepresented backgrounds.

**Synapse to Self Lecture Series**, *Co-Organizer*, Wesleyan University 2006 – 2007  
Co-authored grant proposal as an undergraduate, with two faculty members, for an interdisciplinary lecture series about the mind and brain.

**Mind Matters: Wesleyan Journal of Psychology**, *Senior Editor*, Wesleyan University 2005 – 2008  
Published three annual journals of original undergraduate research in psychology and neuroscience. Organized all aspects of editorial process. Collaborated with other members of editorial board to coordinate the publication process. Maintained website. Recruited and interviewed student reviewers. Arranged and led meetings with student reviewers.

## TEACHING EXPERIENCE

### University of Chicago

|                                                                                            |             |
|--------------------------------------------------------------------------------------------|-------------|
| Mentor, three graduate students from the University of Chicago                             | 2017-2024   |
| Mentor, four undergraduates from University of Chicago and one from Tulane                 | 2014-2024   |
| Teaching assistant, Brains! workshop with local primary school                             | Dec 2015    |
| Participant, Course Design and College Teaching, University of Chicago Center for Teaching | Spring 2014 |
| Teaching assistant, Computational approaches to cognitive neuroscience                     | Spring 2014 |
| Teaching assistant, Mathematical methods in the biological sciences                        | Fall 2013   |

### Wesleyan University

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|---------------------------------------------|-----------|
| Teaching Assistant, Behavioral Neurobiology | Fall 2008 |
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## HONORS AND AWARDS

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|------------------------------------------------------------------------------------|--------------------|
| Travel Award, Neurocybernetics at Scale at the Champalimaud Centre for the Unknown | 2025               |
| Travel Award, 10 Questions in Non-Human Primate Neuroscience at Carnegie Melon     | 2025               |
| NSF Integrative Training in Motor Control and Movement (moto-IGERT)                | 2014-2015          |
| Howard Hughes Summer Fellowship                                                    | Summer 2007        |
| Mind and Life Summer Research Institute Research Fellow                            | Summers 2007, 2008 |
| Wesleyan Center for the Humanities Fellowship                                      | Fall 2007          |
| Edward W. Snowdon Fund Wesleyan                                                    | 2006               |
| Freeman Family Grant for the Study of East Asian Diaspora                          | Summer 2006        |